HOORVIA - MATH & LOGIC SYMBOLS CHART: PRINTABLE REFERENCE GUIDE

Quick Reference Cheat Sheet · A4 Printable

Reference Charts · Sheet #1

Symbol	Name	Meaning	Example
?	For All	Universal quantifier: every member satisfies the condition	?x, x + 0 = x
?	There Exists	Existential quantifier: at least one member satisfies it	?x such that $x^2 = 4$
?	Element Of	Membership: the item belongs to the set	3 ? {1, 2, 3}
?	Not an Element Of	Non-membership: the item is outside the set	5 ? {1, 2, 3}
?	Subset	Every element of the left set is also in the right set	{1, 2} ? {1, 2, 3}
?	Union	All elements from both sets combined	{1, 2} ? {2, 3} = {1, 2, 3}
?	Intersection	Only the elements found in both sets	{1, 2} ? {2, 3} = {2}
?	Logical AND	Conjunction: both statements must be true	p ? q is true only if both hold
?	Logical OR	Disjunction: at least one statement is true	p ? q is true if either holds
¬	Logical NOT	Negation: flips true to false	¬p is true when p is false
->	Implies	Conditional: if the left holds, the right follows	p -> q: p true forces q true
?	If and Only If	Biconditional: both sides always match	p ? q is true when both agree
?	Therefore	Marks the conclusion of an argument	All men are mortal ? Socrates is mortal
?	Summation	Adds a sequence of terms together	?(1 to n) i = n(n+1)/2

PRO TIP: Read Symbols as Words: ?x is read 'for all x' and ?x as 'there exists an x'. Saying the sentence out loud is the fastest way to make a proof readable.